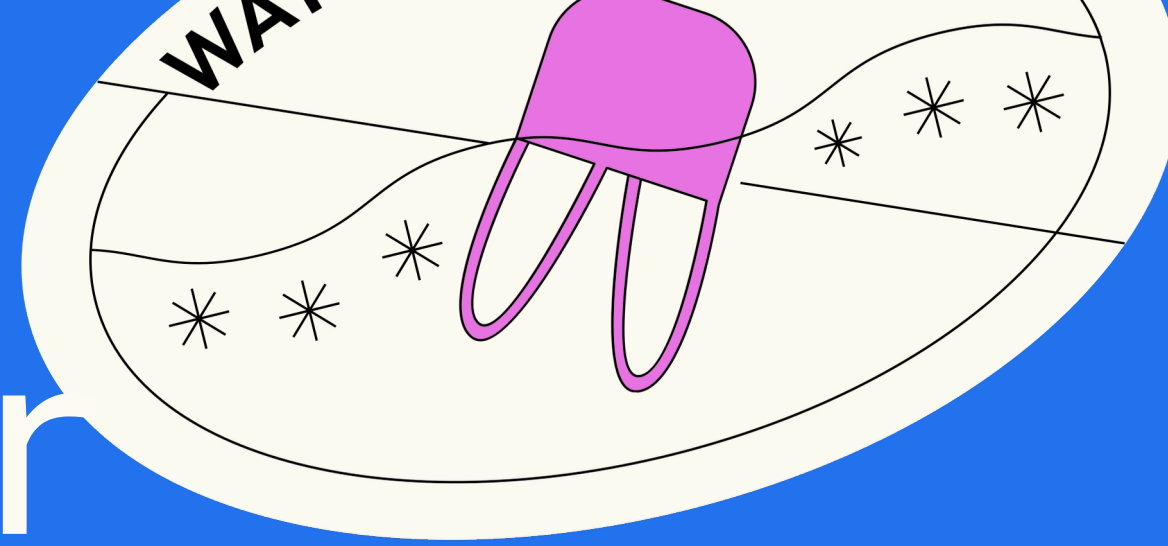
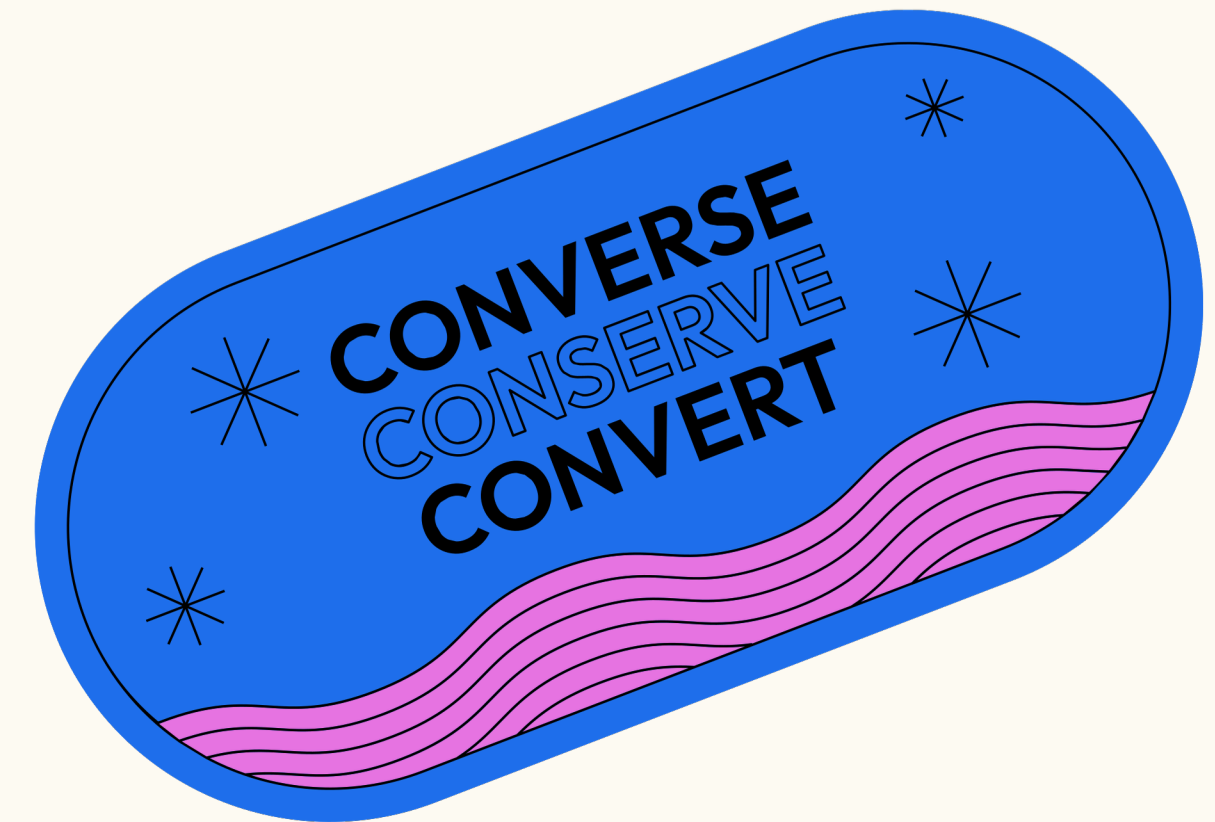
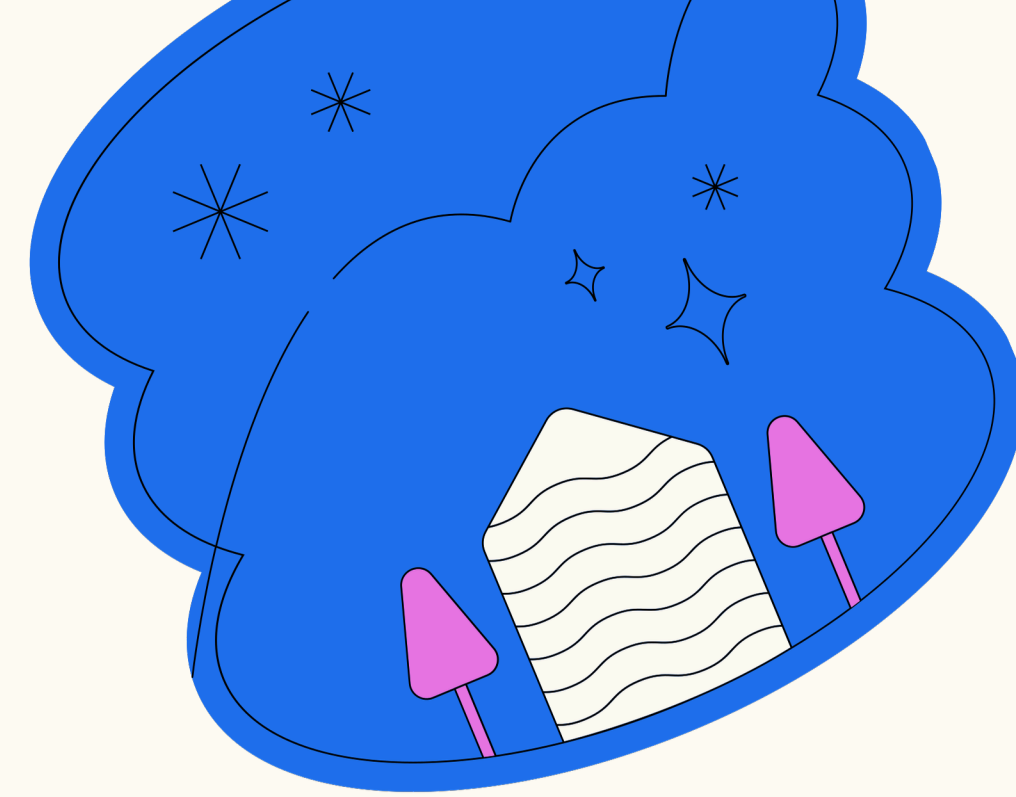


GroundWater Level Prediction using Machine Learning

A Machine Learning
Approach for Sustainable
Water Management



01-Introduction
02-Problem
statement
03-Dataset
04-Methodology
and Models used
05-Performances
06-Applications
07-Conclusion
and References



01 – INTRODUCTION

- Groundwater: vital for agriculture, drinking water, and industry
- Problem: depletion due to over-extraction, climate change, poor management
- Traditional monitoring: slow & location-specific
- Data Science + ML: can predict groundwater levels → helps sustainable management



02 – PROBLEM STATEMENT



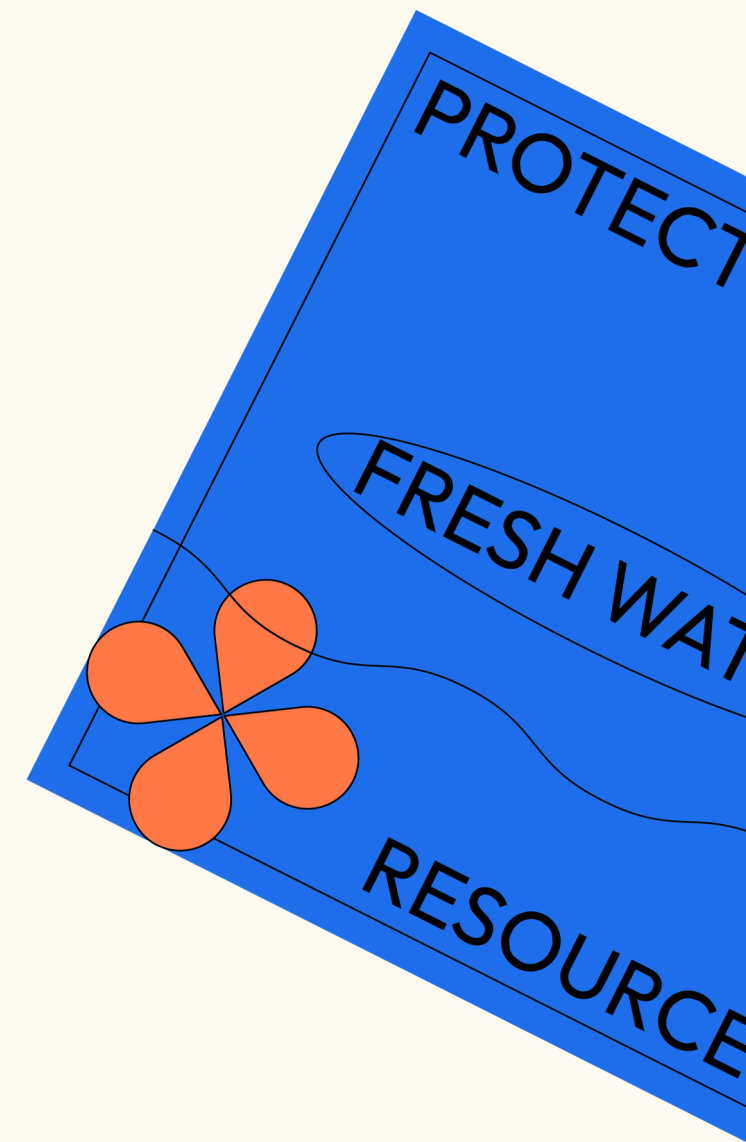
- Declining groundwater threatens agriculture, water supply & ecosystems
- Traditional methods insufficient for climate/rainfall variability
- Objective: Build ML model to predict groundwater depth using historical & climate data
- Impact:
 - Efficient irrigation planning
 - Sustainable water management policies
 - Early warning systems





03 – DATASET

- Primary Source: Central Ground Water Board (CGWB), India
- Features: Well ID, location (lat/long, elevation) Date, Water level
- depth Optional: water quality
- Supplementary Data: Rainfall (IMD), temperature, soil type, cropping pattern
- Frequency: Monthly (can be resampled to daily/weekly)
-

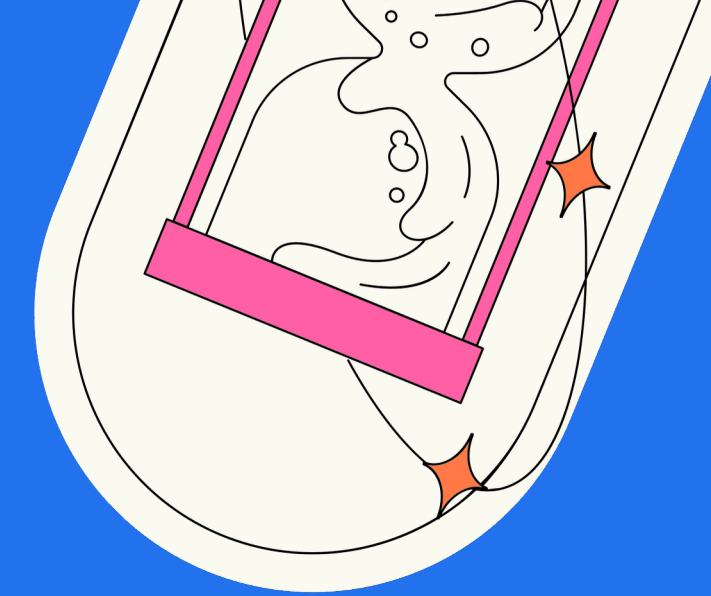


04 – METHODOLOGY AND MODELS USED

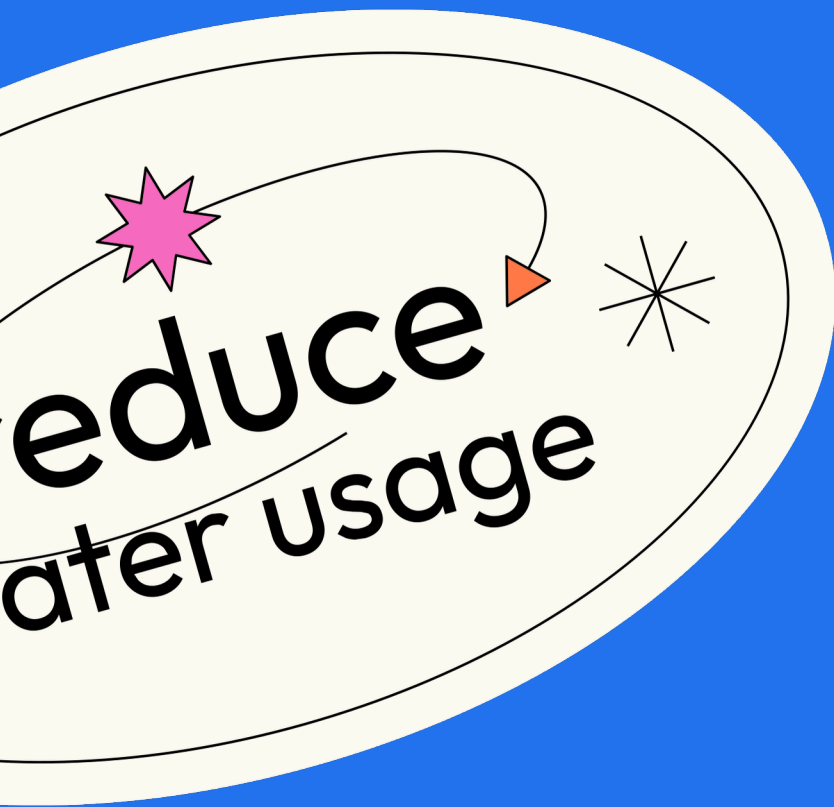
METHODOLOGY

- Data Collection – groundwater + climate data
- Preprocessing – missing values, resampling, outlier removal
- Feature Engineering – lags, rolling stats, rainfall indices, seasonality, soil type
- Modeling – Persistence, ARIMA/SARIMA, Random Forest, XGBoost, LSTM
- Evaluation – RMSE, MAE, R^2 , NSE; time-series split validation

4 – METHODOLOGY AND MODELS USED

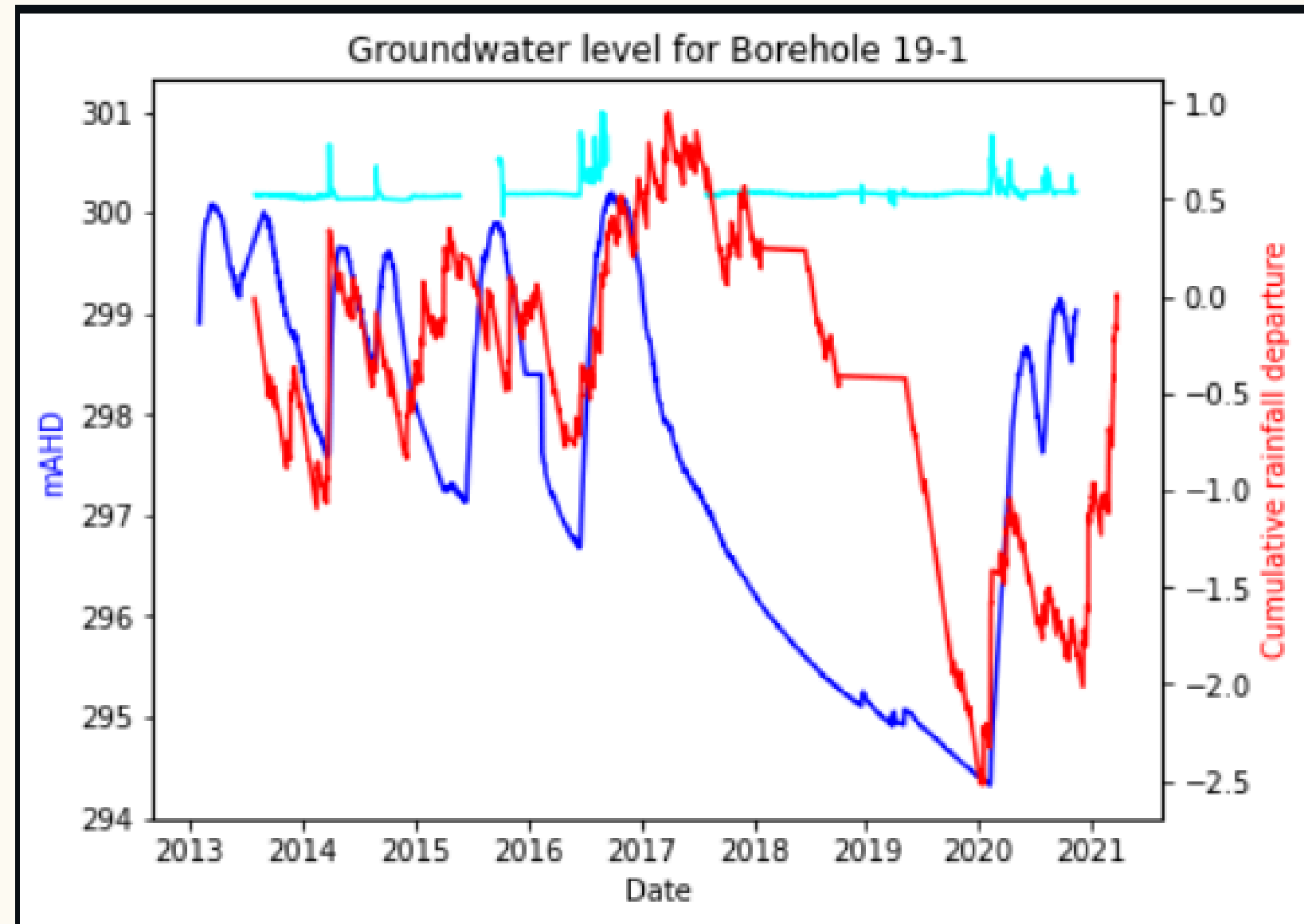


MODELS

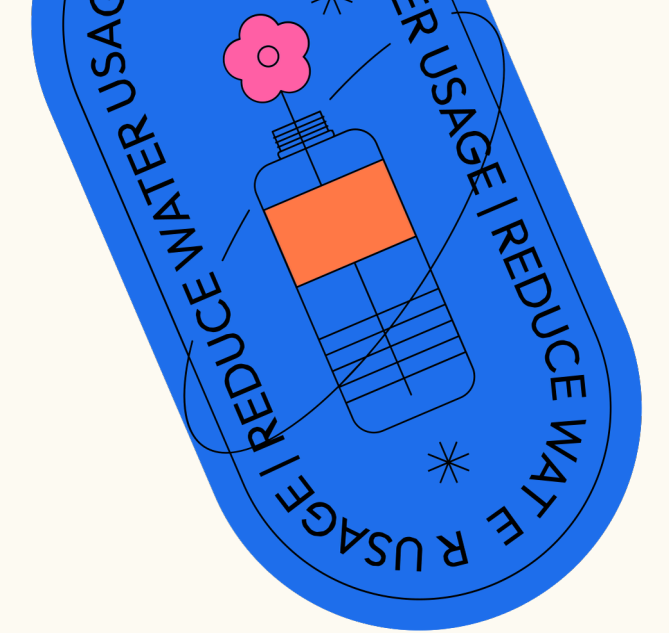
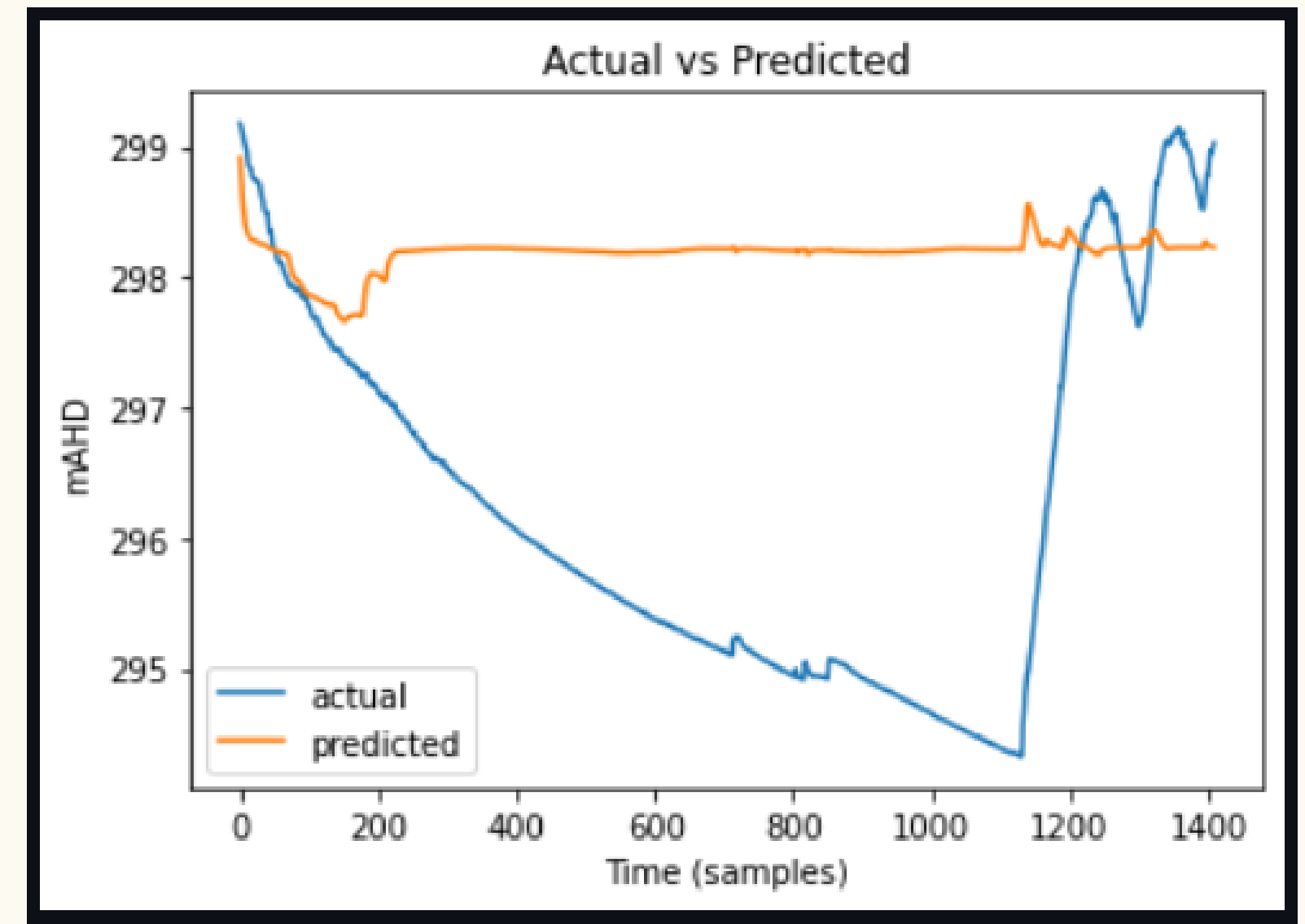


- Baseline Model – Persistence
- Statistical Models – ARIMA/SARIMA
- Machine Learning Models – Random Forest & XGBoost
- Deep Learning Model – LSTM

05 – PERFORMANCE



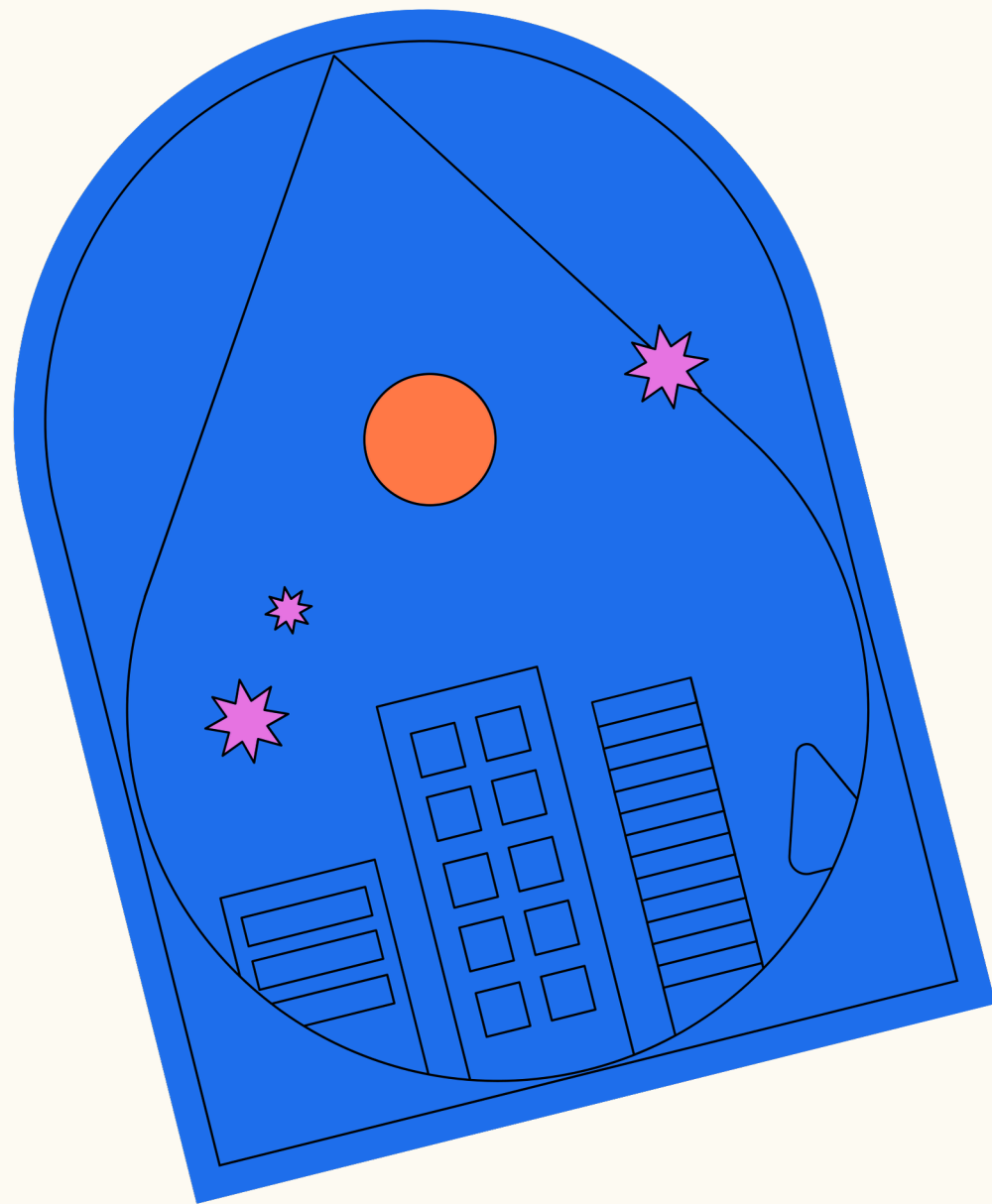
Observation: LSTM best overall;
XGBoost good for short-term &
interpretability



06 – APPLICATIONS



- Early warning systems for water scarcity
- Irrigation planning → optimize water use
- Policy-making support for government
- Smart irrigation with IoT devices



04 – Conclusions AND REFERENCES

CONCLUSION

ML and DL models accurately predict groundwater using time-series and climate data, capturing long-term patterns for effective decision-making. Future work includes integrating climate change scenarios, IoT-based monitoring, and real-time dashboards for improved groundwater management.

REFERENCES

- Central Ground Water Board (CGWB) – <http://cgwb.gov.in> Indian
- Meteorological Department (IMD) – <https://www.imdpune.gov.in> Key research
- papers (Kuhn & Johnson, Hochreiter & Schmidhuber, Pedregosa et al.)



Thanks

